🧪 jQuery Lab Exercises & Assignments

🔹 LEVEL 1 – BASIC LAB EXERCISES (Selectors & DOM)

🧪 Lab 1: jQuery Selectors Practice

🎯 Objective

Understand how jQuery selects elements.

🔧 Tasks

1. Select all <p> elements and change text color to blue.
2. Select element by ID and change background color.
3. Select elements by class and add border.
4. Select first and last list items.
5. Hide all even rows in a table.

💡 Concepts Covered

- $()
- #id
- .class
- :first, :last
- :even, :odd

🧪 Lab 2: DOM Manipulation

🎯 Objective

Modify HTML dynamically.

🔧 Tasks

1. Change text using .text()
2. Change HTML using .html()
3. Change input value using .val()
4. Add new element using .append()
5. Remove element using .remove()

📝 **Assignment**

Create a **Dynamic Todo List**:

- Add task
- Delete task
- Clear all tasks

🔹 **LEVEL 2 – EVENTS & EFFECTS**

🧪 **Lab 3: Event Handling**

🎯 **Objective**

Handle user interactions.

🔧 **Tasks**

1. Show alert on button click.
2. Change image on hover.
3. Double click to hide paragraph.
4. Show message when input loses focus.
5. Prevent form submission.

💡 **Methods**

- .click()
- .dblclick()

- .hover()
- .focus()
- .blur()
- .submit()

🧪 **Lab 4: jQuery Effects**

🎯 **Objective**

Work with animations.

🔧 **Tasks**

1. Hide/Show using .hide() and .show()
2. Toggle visibility.
3. Use .fadeIn() and .fadeOut()
4. Use .slideDown() and .slideUp()
5. Create custom animation using .animate()

🔹 **LEVEL 3 – AJAX & API INTEGRATION**

🧪 **Lab 5: AJAX Basics**

🎯 **Objective**

Call API and display data.

Use this API:

👉 https://jsonplaceholder.typicode.com/users

🔧 **Tasks**

1. Fetch users using $.ajax()
2. Display data in table

3. Show loader before API call
4. Handle error case
5. Add search filter

🧪 Lab 6: CRUD using JSON Server

(Important because you already practiced this)

🎯 Objective

Perform CRUD using jQuery + JSON Server

🔧 Tasks

1. Display all books
2. Add new book
3. Edit book
4. Delete book
5. Handle numeric & string ID issue

📝 Assignment Enhancement

Add:

- Form validation
- Confirmation dialog before delete
- Toast message after success

🔹 LEVEL 4 – MINI PROJECT ASSIGNMENTS

📘 Assignment 1: Student Management System

Features:

- Add Student

- Edit Student
- Delete Student
- Search Student
- Filter by Grade
- Pagination

Concepts:

- jQuery selectors
- Events
- DOM manipulation
- AJAX
- Validation